

F1R3Skein: Drawn Distributions

Weighted state machines for the pitch and duration streams

Feature design, revision 2 — companion to the instrument specification, revision 4

F1R3FLY.io

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Abstract

In the instrument as built, each ribbon is the base- b digit expansion of a transcendental constant. The pitch ribbon runs in base $3d + 1$ — three octaves of a d -degree scale plus a rest — and the duration ribbon in base 5. The distribution of notes is therefore whatever the constant happens to supply. This feature lets the musician, M, *draw* that distribution instead: a weighted state machine over the ribbon’s states, one disc per state arranged in a circle, arrows dragged from disc to disc, each arrow carrying a weight.

The design rests on one decision:

M supplies the distribution; the constant still supplies the chance.

The digits of the constant are used as an exact, lazily-read uniform variate that drives the machine, by interval refinement over the machine’s cumulative weights. Everything the platform already depends on survives unchanged: realisation is deterministic, performances replay exactly, tunes remain terms, $\text{Snip}(k, i_L, i_R, n)$ still addresses them, and breeding still operates on map-agnostic material. The original instrument is recovered exactly as one machine among many (Proposition 2). The editor itself is an all-or-nothing modal window: M commits a valid machine or cancels, with no intermediate save.

Revision 2 records the decisions taken on revision 1 (§10): every recommendation is accepted, the duration vocabulary is fixed as whole note to sixteenth with index 0 the longest, and the evolution of machines is assigned to a later phase.

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1 What was reviewed

The design is written against the v2 branch of F1R3FLY-io/F1R3Games, head 18a675a (15 September 2026), which is where the current implementation lives:

- `design/f1r3skein-instrument-spec-v4.tex` — the specification this feature extends, in particular §3.3 (note mapping), §7 (snip semantics and the envelope), §8 (the performance trace) and §9 (the wire protocol).
- `design/Skein.module` — the Theory of skeins.
- `F1R3Skein/V2/f1r3skein/` — the implementation: `skein-spigot` (digit streams, memoised, bases 2..64), `skein-core` (`term.rs`, `envelope.rs`, `state.rs`), `skein-proto` (protocol v4), `skein-engine`, and the visionOS client under `swift/SkeinClient` (`ControlPanel`, `SoundPanel`, `Scale`, `SkeinModel`).

Four facts from the code shape everything below.

1. **Roles hold bases; only constants move.** `Skein::Weave` carries the pitch constant, the duration constant, the pitch base (16, 22 or 37) and the duration base (always `DURATION_BASE = 5`). A twist exchanges the constants and nothing else.
2. **The top pitch digit is the rest.** `Envelope::render` maps digit $b - 1$ of the pitch stream to silence (`top_digit_is_rest`). It is generated silence: addressed, captured and bred like any note.
3. **Material is index pairs.** `realise` turns a term into `Cells` of (p, d) indices; the envelope alone turns indices into MIDI pitches and ticks. That is what makes breeding map-agnostic.
4. **Stream changes happen only while unzipped.** `Instrument::set_streams` rejects with `AlreadyZipped` once a mesh exists, resets the affected cursor to zero on any change, and the scale picker in `SoundPanel` is disabled while zipped.

2 The machine

2.1 Definition

Fix a role $\rho \in \{\text{pitch}, \text{duration}\}$ with base N_ρ : $N_{\text{pitch}} = 3d+1$ for the selected scale, $N_{\text{duration}} = 5$. The states are the ribbon’s symbols $D_\rho = \{0, \dots, N_\rho - 1\}$, and in the pitch role state $N_\rho - 1$ is the rest.

Definition 1 (Drawn machine). A *drawn machine* for role ρ is a tuple $\mathcal{M} = (N, E, w, \mu)$ where $N = N_\rho$, $E \subseteq D_\rho \times D_\rho$ is the set of drawn arrows (self-loops permitted), $w : E \rightarrow \mathbb{Q}_{>0}$ assigns each arrow a positive rational weight, and μ is the initial distribution: either uniform over the discs of the component, or pinned to one start disc (§5.6).

Write $V(E)$ for the discs incident to at least one arrow — the discs M has actually used. Discs outside $V(E)$ are simply never visited: drawing a machine over the diatonic discs that ignores the top octave removes the top octave from the stream.

2.2 Weights are relative; each row is normalised

M’s requirement is that a new arrow be labelled “as if it were one of a cone to every other state, all transitions equally likely and summing to one”. So the default weight of a new arrow is

$$w_0 = \frac{1}{N - 1},$$

$\frac{1}{21}$ for diatonic, $\frac{1}{15}$ for pentatonic, $\frac{1}{4}$ for durations. We call this the *uniform-cone default* rather than a “normal” weight, to keep it clear of the Gaussian.

M will almost never draw a complete cone, so a row of drawn weights will not in general sum to one: three default arrows out of a diatonic disc sum to $\frac{3}{21}$. Requiring M to balance every row by hand would make the editor an arithmetic exercise. Weights are therefore *relative*, and the transition probability is the row-normalised weight

$$P(s \rightarrow t) = \frac{w(s, t)}{\sum_{(s, t') \in E} w(s, t')}.$$

The editor shows both: the weight M typed on the arrow’s chip, and beneath it the effective probability. The consequence M sees is the natural one: all-default arrows out of a disc are equally likely whatever their number, and doubling one arrow’s weight makes it twice as likely as each of its default siblings. The absolute value of w_0 only sets the scale against which M edits.

2.3 Validity

M must end the activity with either a valid machine or a cancellation. A machine is valid when:

- V1** *Non-empty.* $E \neq \emptyset$.
- V2** *Every arrow weighted.* $w(e) \in \mathbb{Q}_{>0}$ for every $e \in E$. This holds by construction, since an arrow is born carrying w_0 ; the editor refuses zero or negative entries and offers deletion instead, because a zero-weight arrow and an absent arrow are the same machine.
- V3** *One connected component.* The undirected graph on $V(E)$ is connected. This is M’s stated rule.
- V4** *No dead ends.* Every disc in $V(E)$ has at least one outgoing arrow.

V5 *Right size.* N equals the current base of the role. (Relevant only at commit, if the scale changed while the editor was open.)

V4 is an addition to M's stated rule, and needs justifying. Specification §3.1 requires a stream to be unbounded. A disc with arrows in but none out is a place the chain can reach and never leave, so the ribbon would simply end there. Blocking commit and highlighting the offending discs keeps the semantics explicit. *Settled:* dead ends block commit; there is no implicit return to the start distribution.

Two further conditions are *reported but not enforced*, because they are musical choices rather than errors. If the machine is not *strongly* connected, the chain can enter a closed set of discs and stay there; the editor shades the transient discs so M knows the opening material will not return. That is often the intent — an introduction that falls into an ostinato. And if the only cycles have a common period (a strict alternation, say), the editor notes it.

3 Realisation: the constant as dice

3.1 Why the constant stays

A machine alone does not determine a stream; it needs a source of chance. Three facts decide what that source must be. Performances replay exactly from their trace (spec §8). A tune is a term whose realisation is a pure rewrite (spec §7). And the research note requires a tune realised on a player's machine to be bit-identical to the same tune realised by a validator. A seeded PRNG would satisfy the first two; floating-point sampling would fail the third. The constant already present in the role satisfies all three, with exact integer arithmetic — and it keeps the instrument's identity: the ribbon is still π , now heard through M's machine.

3.2 Exact interval decoding

Let the role's configuration be constant c in base $b = N_\rho$, with digits $\delta_0\delta_1\dots$ of $\text{frac}(c)$ as *skein-spigot* emits them. At driver position q the unread tail denotes the real number $x_q = \text{frac}(b^q c) \in (0, 1)$. For a row with successors $t_1 < \dots < t_m$ and cumulative probabilities $0 = C_0 < C_1 < \dots < C_m = 1$, decoding reads digits one at a time, narrowing x_q to a b -adic interval, until that interval lies inside a single bin $[C_{j-1}, C_j]$.

Algorithm 1 $\text{Dec}(\text{row}, q)$ — one transition, exact

```

1:  $A \leftarrow 0$ ;  $k \leftarrow 0$   $\triangleright$  interval is  $[A/b^k, (A+1)/b^k]$ 
2: repeat
3:    $A \leftarrow A \cdot b + \delta_{q+k}$ ;  $k \leftarrow k + 1$ 
4:    $j \leftarrow$  the index with  $C_{j-1} b^k \leq A$  and  $(A+1) \leq C_j b^k$ , if one exists
5: until  $j$  found or  $k = K_{\max}$ 
6: if no  $j$  then
7:    $j \leftarrow \max\{i : C_{i-1} b^k \leq A\}$   $\triangleright$  deterministic fallback
8: end if
9: return  $(t_j, q + k)$   $\triangleright$  next state, advanced driver position

```

All comparisons are integer comparisons after clearing the row's common denominator; there is no floating point anywhere. The symbol stream of role ρ is then

$$\sigma_0 = \text{Dec}(\mu, 0)_1, \quad (\sigma_{t+1}, q_{t+1}) = \text{Dec}(\text{row}(\sigma_t), q_t),$$

with q_0 the digits consumed drawing σ_0 (none when μ is pinned). Notch t of the ribbon displays, and meshes, σ_t .

Proposition 1 (Termination). *For irrational c , the loop of Algorithm 1 terminates without the fallback.*

Proof. $x_q = \text{frac}(b^q c)$ is irrational because c is, so it equals none of the rational boundaries C_j . It therefore lies strictly inside its bin, at some distance $\varepsilon > 0$ from both ends. After k digits the interval $[A/b^k, (A+1)/b^k)$ contains x_q and has width b^{-k} ; once $b^{-k} < \varepsilon$ it lies inside the bin and the loop exits. $\square$

All five constants are irrational, so $K_{\max}$ (proposed: 64) is a guard against pathological inputs, not a rounding rule. It is recorded in the configuration so that it forms part of the definition of realisation.

3.3 The original instrument is one machine

Proposition 2 (Embedding). *Let $\mathcal{U}_N$ be the complete machine on N discs with every arrow including every self-loop drawn at the default weight, and μ uniform. With $b = N$, the symbol stream of $\mathcal{U}_N$ is exactly the digit stream: $\sigma_t = \delta_t$ for all t .*

Proof. Every row normalises to $1/N$, so the bins are $[j/N, (j+1)/N)$. With $b = N$ the first digit read places the interval $[A/N, (A+1)/N)$ exactly on bin A , so each draw consumes one digit and returns it. $\square$

So the feature does not replace the old design; it generalises it. Every existing tune is a tune of $\mathcal{U}_N$, and the spigot-only code path can remain as the fast path for that case.

3.4 Measured behaviour

A Python prototype of Algorithm 1 (exact rationals, 6000–8000 transitions) gives:

Machine	Driver	digits/notch	max	observed
5 discs, mixed rows	π , base 5	1.38	6	from disc 0 (three arrows, $\frac{1}{3}$ each): .338 .357 .305
same	e , base 5	1.35	6	.325 .331 .344
same	Liouville, base 5	1.00	2	1.000 to the first arrow, 0 to the others
22-disc full cone	π , base 22	1.96	5	state frequencies .040–.051 against $\frac{1}{22} = .045$

Two things follow. Decoding is cheap: under two digits per notch on average, so realisation cost stays within a small constant of the spigot’s. And **the constant still colours the result wherever it departs from normality**. π , e , φ and $\ln 2$ are conjectured, not proved, normal; empirically they behave as uniform dice. Liouville’s constant is provably not normal — its digits are almost all zero — so it drives every machine to the lowest bin of every row, turning a drawn machine into a deterministic walk along first arrows. That is a legitimate and rather striking instrument, but it should be a known one. *Settled*: drawn machines are permitted on Liouville’s constant, and the editor MUST say what it will do whenever the role’s constant is Liouville’s.

3.5 Cost and memoisation

A drawn stream is sequential: σ_t depends on the whole prefix. The engine keeps, per (configuration, machine), a vector of symbols and a sparse table of checkpoints (t, σ_t, q_t) every 256 notches, so re-realising a visited range is $O(1)$ per notch as spec §3.1 requires, and a first visit to depth t costs $O(t)$ decoder steps over already-memoised digits. This is the resource floor the research note wants: deep addresses cost more than shallow ones.

4 Where the machine lives

4.1 In the term, not the envelope

The envelope is *interpretation*: given indices, which pitches and which ticks. The machine decides *which indices occur*. The same $\text{Snip}(k, i_L, i_R, n)$ read through two machines yields different material. The machine is therefore material, and belongs in the skein:

```
pub enum Skein {
  Weave {
    pitch: Constant, duration: Constant,
    pitch_base: u32, duration_base: u32,
    pitch_machine: Option<Machine>, // None: the plain spigot
    duration_machine: Option<Machine>,
  },
  Twist(Box<Skein>),
}
```

This keeps each existing law intact.

- **Twist.** Machines are sized by their role, so they stay with the role, like the bases:

$$\text{Twist}(Wv\ p\ d\ p_b\ d_b\ p_m\ d_m) = Wv\ d\ p\ p_b\ d_b\ p_m\ d_m.$$

After a twist, e drives the pitch machine and π the duration machine. The involution still follows from the one equation.

- **Snip.** Unchanged in form. i_L, i_R index notch positions t , not driver digits.
- **Material and breeding.** `realise` replaces `cache.range(cfg, i, n)` with `symbols(role, i, n)`; Cells are still (p, d) pairs. XPitch, XDur, Pad and the mutations operate on material and need no change, so crossing a tune drawn from a machine with a tune drawn from a bare spigot is already defined.
- **Envelope.** Unchanged. The rest is still applied at render time, so a pitch machine that makes state $N - 1$ likely is a machine that phrases with silence.

4.2 Canonical form

Tunes are de-duplicated by normalisation, so the machine is stored canonically: arrows sorted by (source, target), each row normalised to reduced rationals, μ recorded as `Uniform` or `Pinned(s)`. Two drawings that differ only by scaling a row are the same distribution and the same term. Inline storage is right for `skein/v0`; on the shard a machine becomes a content-addressed value referenced by digest, since one machine will head many tunes.

4.3 Scale changes

A pitch machine is a machine on $3d + 1$ discs. Changing between maps of equal cardinality — major, minor, Dorian — changes the envelope only, and the machine survives: the discs relabel and the material is untouched. Changing cardinality (pentatonic to diatonic) invalidates it. The engine MUST then drop the pitch machine along with resetting the cursor, as `set_streams` already does for a base change, and the panel MUST ask before doing so, naming the machine that will be lost. Nothing is kept in reserve; there are no intermediate saves in this version.

4.4 The Theory

`Skein.module` gains a `Machines` theory. The arrows of a machine are unordered, so they are a `HashBag` — the collection sort the frozen grammar already supplies, unlike the `List` that `Seq` still awaits.

```

Theory Machines(k: Counting) {
  k
  Types { Weight; Arrow; Init; Machine; }
  Exports { Machine; }
  Terms {
    Ratio . p:Nat, q:Nat          |- p "/" q          : Weight;
    Arr   . s:Nat, t:Nat, w:Weight |- s "->" t "@" w    : Arrow;
    Unif  .                       |- "uniform"         : Init;
    Pin   . s:Nat                 |- "start" s          : Init;
    Mach  . n:Nat, i:Init, es:HashBag(Arrow)
          |- "machine" n i "{" es.*sep(",") "}"         : Machine;
  }
}

```

Wv then takes two `Option(Machine)` arguments and the twist equation is restated as above. Row normalisation is a canonicalisation performed by the normaliser, not an equation, since it is not expressible as a finite set of rewrite rules over `Nat` without division. One pre-existing discrepancy should be fixed at the same time: the module's `Weave` still swaps whole streams under `Twist`, whereas `skein/v0` swaps constants only (§7).

5 The editor

5.1 Entry and exit

The `streams` section at the top of `ControlPanel` gains two buttons: **Pitch distribution...** and **Duration distribution...** Each opens a third window, `WindowGroup(id: "machine", for: Role.self)`, alongside the panel and the sound window; one editor at a time. Under each ribbon label the panel shows its source: “ π base 22” or “ π base 22 · drawn (9 discs, 14 arrows)”.

The editor is modal in the sense M asked for.

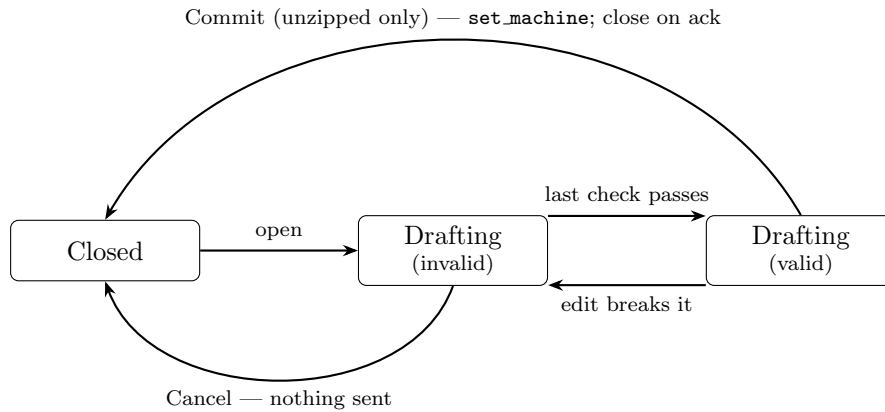


Figure 1: Editor lifecycle. Cancel is available in both drafting states; Commit only when valid and unzipped.

- **Commit** is enabled only when V1–V5 hold *and* the instrument is unzipped. It sends one `set_machine` message; the window closes on the engine’s acknowledgement and stays open, with the reason shown, on a rejection.
- **Cancel** discards the draft and sends nothing. The engine never sees an unfinished machine, which is the whole of the “no intermediate save points” rule.

- The editor MAY be opened while zipped. M can draft while the wave runs; the Commit button says “unzip to apply”, mirroring the scale picker.
- Opening seeds the canvas with the role’s current machine, if any, so an edit is a revision rather than a redraw. A **Clear** control empties the canvas, and **Use the constant** commits None, returning the role to the bare spigot.

5.2 Layout

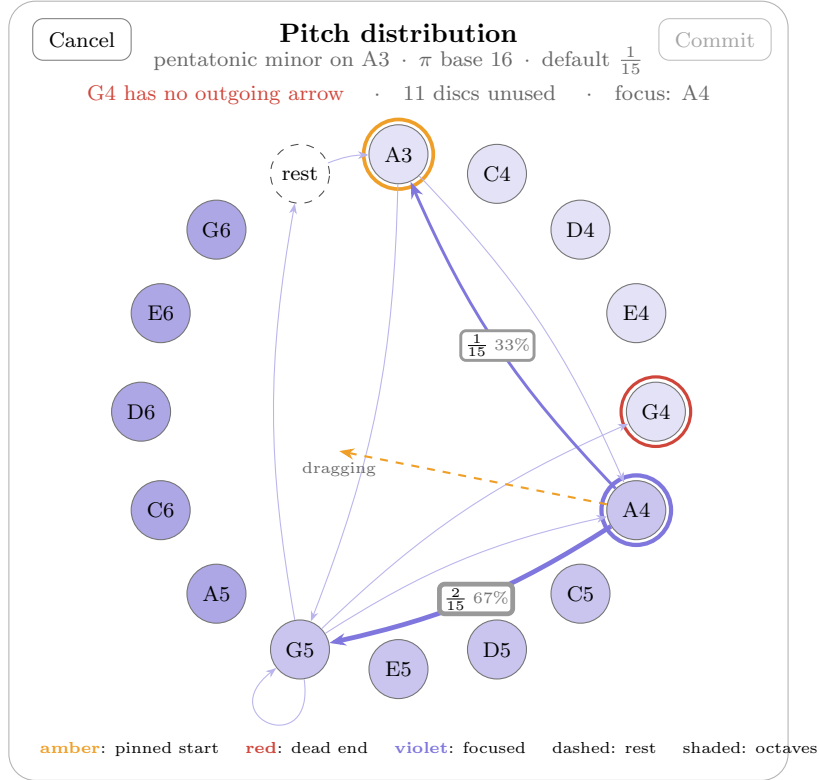


Figure 2: The pitch editor for pentatonic minor: sixteen discs, fifteen pitches over three octaves plus the rest. A4 is focused, so its two arrows carry chips showing the weight as entered and the effective probability; every other arrow is a hairline. The draft is invalid — G4 has arrows in but none out — so Commit is disabled and the reason is stated at the top, where it is legible.

Discs are placed clockwise from twelve o’clock in index order, so pitch rises around the circle and the rest sits just before the lowest note. Labels come from the envelope in force (root and map), so the same machine relabels when M switches between equal-cardinality maps. Octaves are shaded; the rest disc is hollow and dashed, since it is the one disc that is not a pitch.

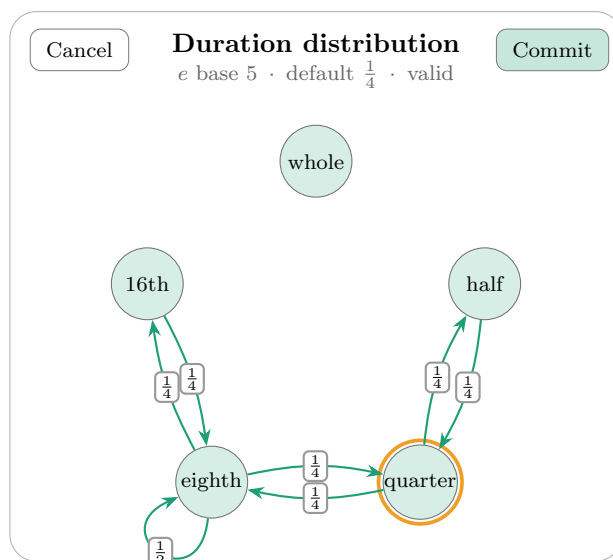


Figure 3: The duration editor: five discs. This machine lives mostly on quarters and eighths, lingers on eighths (the self-loop at $\frac{1}{2}$ against two $\frac{1}{4}$ siblings gives probability $\frac{1}{2}$), and never plays a whole note. Valid, so Commit is live.

5.3 Interactions

Draw an arrow

Look at a source disc and pinch; move the pinched hand. A rubber-band arrow follows the hand's projected position on the window plane. Releasing over a disc (snap radius one disc diameter) creates the arrow with weight w_0 ; releasing over empty space creates nothing.

Self-loop

Drag out beyond the disc and back onto it. A minimum excursion distinguishes this from an accidental pinch-and-release, which only focuses the disc.

Duplicate

Drawing an arrow that already exists does not create a second one; it opens the existing arrow's weight editor. E is a set.

Edit a weight

Tap the arrow's chip. A popover offers a fraction or decimal field ($2/15$, 0.25), steppers $\times 2$ and $\div 2$, **Reset to w_0** , and **Delete arrow**. Entry is converted to a reduced rational immediately; zero and negative entries are refused with the suggestion to delete. The popover also shows the whole row's resulting probabilities, since changing one weight changes its siblings' share.

Focus

Tapping a disc focuses it: its outgoing arrows and their chips come forward, every other arrow fades to a hairline whose thickness still encodes probability. With 22 or 37 discs a full drawing can hold hundreds of arrows, and chips for all of them cannot be legible at once.

Start disc

See §5.6. A long pinch on a disc pins it as the start (amber ring); the same on the pinned disc unpins it.

Undo

Within the draft only. Undo is not a save point; it disappears with the window.

5.4 Validation feedback

After every edit the editor asks the engine to check the draft (§6) and paints the answer: dead ends ringed in red, discs outside the main component dimmed, transient discs hatched, and the first blocking problem stated in words in the header. When the machine is valid the header shows the *stationary distribution* as a thin bar inside each disc, so M sees which notes the machine will favour over a long run — the most useful single fact about a drawn distribution, and one that is not obvious from the weights.

5.5 Ergonomics on the headset

The sound window already moved its dismiss control to the top because gaze tracking is least reliable low in a window, and the September testing found the panel’s footer buttons hard to select. The editor follows that lesson: Cancel and Commit sit in the top bar, as do the validation messages. Discs are at least 60 pt, the visionOS minimum target; for chromatic, 37 discs of 60 pt with spacing need a ring of radius about 450 pt, so the window’s default size is 1000×1000 pt and the ring scales down with it. Chips are also 60 pt targets, which is a second reason chips are shown only for the focused disc.

5.6 The start disc

The initial distribution μ defaults to uniform over $V(E)$; the first symbol is then drawn from the constant like every other. That default is what makes Proposition 2 exact. M may instead pin a start disc — to begin on the tonic, say — in which case no digits are consumed for σ_0 . Pinning is optional and never a validity condition.

6 Engine and protocol

6.1 Core

```
// skein-core/src/machine.rs
pub struct Ratio { pub num: u32, pub den: u32 }           // reduced, > 0
pub struct Arrow { pub from: u8, pub to: u8, pub weight: Ratio }
pub enum Init { Uniform, Pinned(u8) }
pub struct Machine { pub states: u32, pub init: Init, pub arrows: Vec<Arrow> } // canonical

impl Machine {
    pub fn canonical(self) -> Machine;                    // sort, normalise rows, reduce
    pub fn check(&self, base: u32) -> Report;              // V1..V5 + warnings + stationary
}
pub struct ChainStream { /* cfg, machine, symbols, checkpoints */ }
impl ChainStream { pub fn range(&mut self, cache: &mut DigitCache, from: usize, n: usize) ->
    Vec<u8>; }
```

Validation lives in `skein-core`, for the reason the implementation README gives for putting the gesture detectors there: one implementation, unit-testable, shared by configurations S1 and S2. The client does not re-implement V1–V5.

6.2 Messages (protocol v5)

Type	Fields	Note
<i>client</i> → <i>engine</i>		
<code>check_machine</code>	<code>role, machine, seq</code>	side-effect free; debounced to 10 Hz while M edits
<code>set_machine</code>	<code>role, machine</code> or <code>null</code>	<code>null</code> returns the role to the bare spigot
<i>engine</i> → <i>client</i>		
<code>machine_report</code>	<code>role, seq, valid, problems, warnings, stationary</code>	answers a check; stale <code>seq</code> ignored
<code>machine_ack</code>	<code>role, digest</code>	commit accepted; cursor reset to 0
<code>rejected</code>	<code>reason: already_zipped machine_invalid</code>	existing message, new reason
<code>state</code>	<code>+pitch_source, duration_source</code>	labels and digests

`digits` is unchanged in shape; its values are now notch symbols σ_t , which coincide with digits for a bare spigot. `set_machine` is accepted only while unzipped, like every stream change, and resets the role’s cursor to zero because the same position now names a different sequence.

6.3 Trace

Every accepted `set_machine` is a trace record carrying the canonical machine, and the opening declaration of a trace includes both roles’ sources. Without these the trace cannot replay a session, which spec §8 requires. Checks are not recorded; nothing M did in an editor she cancelled reaches the trace.

7 Findings from the review

Three pre-existing issues touch this feature and should be resolved with it.

1. **setScale resets the pitch constant to π .** `SkeinModel.setScale` sends `configure` with `pitchConstant: "pi"` hard-coded. After a twist the pitch role holds e ; changing scale then silently puts π back and zeroes the cursor. With machines this matters more, because a scale change will also be the event that drops a pitch machine. The panel should send the constant the engine reports, or omit it.
2. **The Theory and the code disagree about twist.** `Skein.module` states $(\text{Twist } (Wv\ l\ r)) = (Wv\ r\ l)$ over whole streams; `term.rs` swaps constants and keeps role bases. The code is the settled design (“roles hold the bases; only the constants move”), and the machine extension depends on it, so the module should be brought into line.
3. **The musical duration map is not the one intended.** `DurationMap::Musical` is thirty-second, sixteenth, eighth, quarter, half, in that index order. *Settled*: the musical vocabulary is whole note to sixteenth, with index 0 the longest:

index	0	1	2	3	4
value	whole	half	quarter	eighth	sixteenth
ticks (480 per quarter)	1920	960	480	240	120

`Musical` is changed in place rather than joined by a new map, so the four maps the specification requires (§3.3) are unchanged in name. Because a trace records the map by name, a trace written under protocol v4 would replay with the new values; the engine therefore keeps the old table as a legacy entry selected by the trace’s protocol version, so v4 traces still replay exactly. Disc labels in the duration editor are taken from this table.

One consequence to note, not a decision: `Exponential` ($60 \cdot 2^i$) still runs from shortest to longest. If “index 0 is the longest” is meant to hold for every duration map, it should be reversed in the same change.

Separately, the specification’s base range (2–36) trails the code (2–64), which base 37 for chromatic needs; specification revision 5 should adopt the code’s range.

8 Conformance

1. $\mathcal{U}_N$ with uniform μ realises to the digit stream, for $N \in \{5, 16, 22, 37\}$ and every constant (Proposition 2).
2. A capture taken under a drawn machine re-realises to what was heard.
3. Realisation is bit-identical across runs, processes and configurations S1 and S2.
4. Twist swaps constants and leaves both machines with their roles; twist is an involution.
5. Canonical form: row-proportional drawings are equal terms; arrow order does not matter.
6. `check` rejects each of: empty, disconnected, dead end, non-positive weight, wrong size; and accepts a non-strongly-connected machine with a warning.
7. `set_machine` is rejected while zipped and resets the cursor when accepted.
8. A same-cardinality map change keeps the pitch machine; a cardinality change drops it.
9. Cancel produces no message other than checks, and no trace record.
10. Statistical (not a proof): over 10^5 transitions driven by π , e , φ and $\ln 2$, observed transition frequencies fall within a χ^2 tolerance of the machine’s. Liouville is characterised separately: it always takes the first arrow of each row.
11. Decoder termination: no call reaches $K_{\max}$ over the test corpus.
12. `Musical` maps indices 0..4 to 1920, 960, 480, 240, 120 ticks, and a v4 trace replays under the legacy table to what was heard.

9 Plan of work

D1 — Core.

`machine.rs`: types, canonicalisation, `check`, Algorithm 1, `ChainStream` with checkpoints; extend `Skein::Weave`; route `realise` and `Instrument::ribbon` through symbols; the new `Musical` table with its legacy entry (finding 3). Conformance items 1–6, 10–12. Fully testable without a headset.

D2 — Engine and protocol.

v5 messages, cursor reset, scale-change rule, trace records. Items 7–9. Fix finding 1.

D3 — Editor.

The `machine` window, discs, drag, chips, focus, popover, validation paint, stationary bars. Brought up in the Simulator with pointer input first, as the panel was.

D4 — On device.

Pinch-drag accuracy, snap radius and self-loop excursion tuned on the Vision Pro; the 37-disc chromatic ring is the stress case.

D5 — Theory.

Machines in `Skein.module`; twist equation brought into line (finding 2). Independent of D2–D4.

10 Decisions settled

Revision 1 asked for eight decisions. All recommendations were accepted; the duration vocabulary was fixed; and the evolution of machines was deferred.

Question	Decision
Dead ends	Block commit (V4). No implicit return to μ .
Start	μ uniform over $V(E)$ by default; M may pin a start disc (§5.6).
Self-loops	Permitted. The uniform cone counts every <i>other</i> state, so $w_0 = 1/(N - 1)$.
Duration vocabulary	Whole note to sixteenth; index 0 is the longest. Musical changed in place, with a legacy table for v4 traces (§7, finding 3).
Seeding	The editor opens on the role’s current machine; Clear starts blank.
Driver	Exact lazy decoding, Algorithm 1.
Liouville	Permitted, with a notice in the editor that it walks first arrows.
Machines as breeding material	Out of scope for this feature; a later phase.

Deferred: evolving distributions

Machines are terms, so they can be bred as tunes are — crossover taking rows from each parent, mutation perturbing weights — letting the population evolve distributions as well as melodies. Nothing in this design forecloses that: machines are canonical, content-addressable, and carried in the skein. It is recorded here so that the later phase starts from the representation already chosen.